

MiniCAS Specifications

Furnish and install one Flygt MiniCAS (Mini Control and Status) module to monitor the temperature and leakage detectors installed in each Flygt pump or mixer. The MiniCAS shall be capable of monitoring the thermal switches embedded in the stator end coils, the Flygt FLS (float switch type) water-in-stator-housing sensor, ~~and the Flygt GLS (capacitive type) water-in-oil sensor.~~ The MiniCAS shall monitor both the series connected thermal switches and leakage sensor(s) by outputting 12 VDC on a single two wire circuit. ~~When both GLS and FLS leakage sensors are specified they shall be connected in parallel with each other and then in series with the thermal switches.~~

The MiniCAS circuitry shall operate on the current sensing principle whereby a change in temperature or leakage condition shall change the resistance of the associated sensor and thus alter the current in the sensing circuit. The MiniCAS shall contain two sets of form C dry contacts, one for overtemperature and one for leakage. The dry

contacts shall change status upon occurrence of an over temperature or leakage condition so as to indicate that condition to other control components in the pump control panel. In the case of an overtemperature, and in keeping with Flygt's warranty policy, the overtemperature dry contacts shall be used to trip the pump off line. The MiniCAS shall be designed to be plugged into a standard 11-pin circular socket. Detailed technical data and wiring connections shall be found in the MiniCAS Manual.

Note: 11-pin circular socket by panel provider.

**Typical of Two (2): One (1) provided per Pump.
For mounting and insertion by others in panel by others.**

MiniCAS 120

Features

- Plug in Replacement for MiniCAS/FUS unit
- 120 VAC, 24 VAC, or 24 VDC Powered
- Durable Plastic Enclosure with Flange for Mounting
- Leakage & Temperature Alarm Indication
- Power Applied Indication
- Temperature Alarm Reset Mode Select Switch
- Temperature Alarm Reset Push-button
- Sensor Input Transient & Short Circuit Protected



UL FILE # E132545



Operation

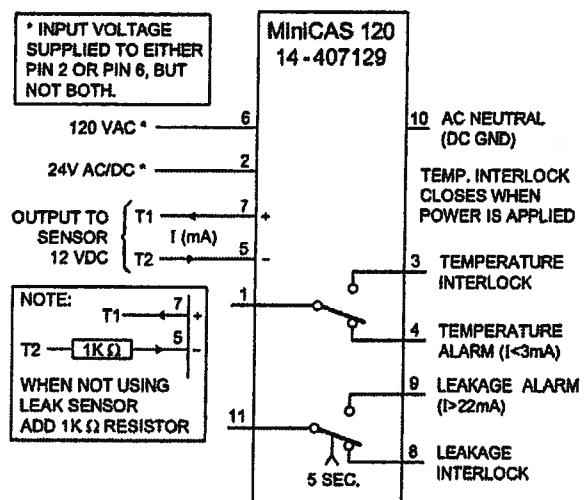
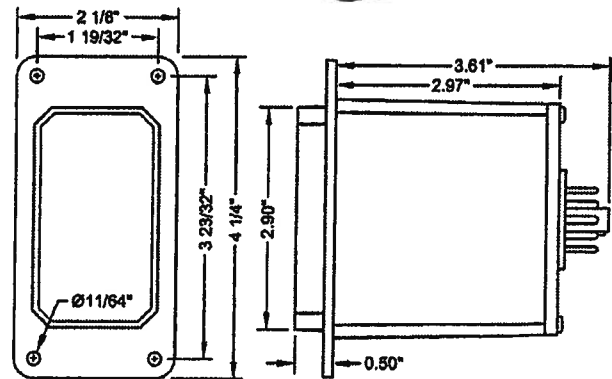
The MiniCAS 120 provides Motor Over Temperature and Seal Leakage alarms for Flygt Submersible Pumps equipped with FLS or CLS sensors.

The unit applies 12 VDC to the sensor and measures the current flow through the sensor. The sensor controls the current in the circuit. If the sensor current is in the normal range the Temperature Alarm Relay is energized to allow normal pump operation. If the sensor circuit becomes shorted, the 12 VDC is turned off and all LEDs flash. Upon a High Motor Temperature condition, the sensor opens so that the sensor circuit current drops to zero. With the sensor current below the Trip Point ($\leq 3.0 \text{ mA} \pm 5\%$), the Overtemp Indication is turned on and the Temperature Alarm Relay is de-energized, preventing pump operation.

When the High Motor Temperature condition has cleared, the unit will reset based on the position of Alarm Reset Mode Select Switch (Auto or Manual). When in the Auto position, the Overtemp Alarm resets automatically. If the switch is in the Manual position, the Overtemp Reset Push-button must be pushed to clear the alarm.

Upon a Seal Leakage condition, the sensor increases the sensor circuit current above the Trip Point ($\geq 22 \text{ mA} \pm 5\%$), the Leakage Indication is turned on and the Leakage Alarm Relay is energized.

MADE IN THE
U.S.A.



Specifications

Input Power:	120 VAC $\pm 10\%$, 7.0 VA max 24 VAC $\pm 10\%$, 3.5 VA max 24 VDC $\pm 10\%$, 125 mA max
Output Rating:	10A Resistive @ 120VAC
Operating Temp:	-20°C to +65 °C
Storage Temp:	-45°C to +85 °C
Sensor Circuit Voltage:	12 VDC $\pm 10\%$
Temp Alarm Trip Point:	$\leq 3.0 \text{ mA} \pm 5\%$
Leak Alarm Trip Point:	$\geq 22 \text{ mA} \pm 5\%$
Enclosure:	Blue Lexan
Base:	Phenolic

ORDERING INFORMATION

Part Number – 14-407129



Issued: 9/11 Supersedes: 6/06

MiniCAS II/FUS-120 Technical Data

MiniCAS II/FUS-120 Technical Data:

Operation Principle: Current sensing

Environment: -20 to 65°C (-4 to 149°F)

Supply Voltage: 120 VAC 50-60 Hz $\pm 10\%$, 24 VAC $\pm 10\%$, 24 VDC $\pm 10\%$

Relay Contact Rating: 10 Amps @ 120 VAC

Voltage to Sensor: 12 VDC $\pm 10\%$

Values of Operation: 3.0 mA $< I < 22$ mA = OK conditions.
 $I \leq 3.0$ mA = High temp. $\pm 5\%$ (or interrupt).
 $I \geq 22.0$ mA = Leakage $\pm 5\%$ (or short circuit).
 (I = current measured by the MiniCAS II/FUS).

 Green LED On = Supply Voltage present.
 Green LED Off = No Supply Voltage present.

Leakage

Contact: Form "C" 10 A @ 120 VAC (N.C. contact for interlocking)

Reset: Automatic (N.O. contact for alarm)

LED Indicators: Red LED On = Leakage Indicated
 Red LED Off = No leakage indicated

Temperature

Contact: Form "C" 10 A @ 120 VAC (N.C. contact for interlocking, N.O. contact for alarm)

Reset: Manual - by interrupting the supply for 1 sec. or by setting the toggle switch in the "Manual" mode.
 Automatic - by setting the toggle switch in the "Auto Reset" mode.

LED Indicators: Red LED On = Over-temperature Indicated.
 Red LED Off = No Over-temperature indicated

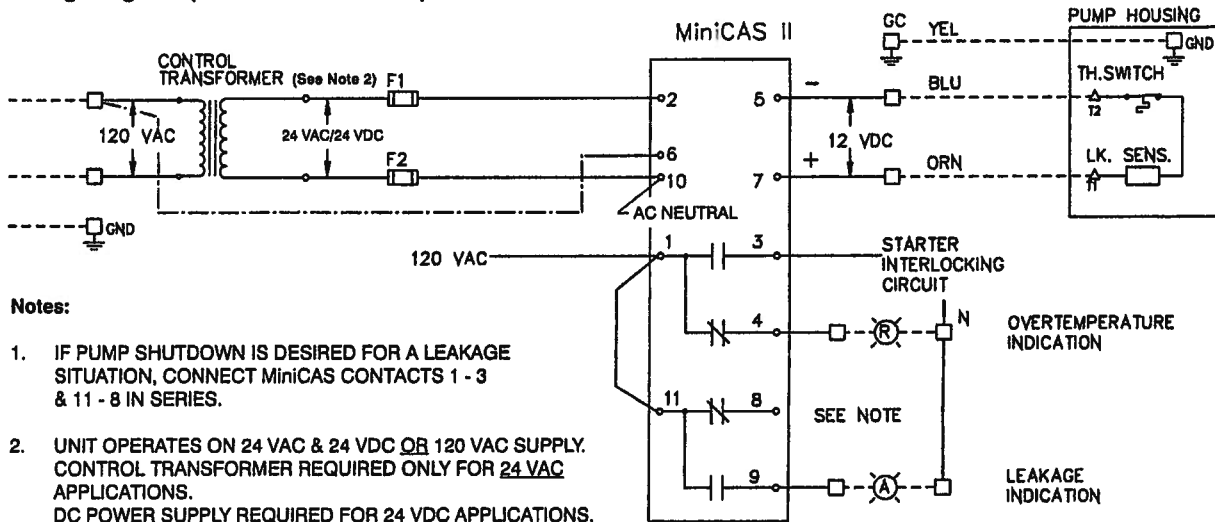
Physical Size: Width: 2.125"
 Height: 4.250"
 Depth: 3.470" (+ socket depth)

Part Number: 14-40 71 29 (MiniCAS II/FUS -120)
 ~~14-40 70 97 (Socket, 11-pin) - optional~~

Approvals: UL - File E101681

Wiring Diagram MiniCAS II/FUS-120

Wiring Diagram (MiniCAS II/FUS-120)



Notes:

1. IF PUMP SHUTDOWN IS DESIRED FOR A LEAKAGE SITUATION, CONNECT MiniCAS CONTACTS 1 - 3 & 11 - 8 IN SERIES.
2. UNIT OPERATES ON 24 VAC & 24 VDC OR 120 VAC SUPPLY. CONTROL TRANSFORMER REQUIRED ONLY FOR 24 VAC APPLICATIONS. DC POWER SUPPLY REQUIRED FOR 24 VDC APPLICATIONS.

Mode of Operation

In normal conditions, when the MiniCAS - 120 is powered, the green LED is 'ON' and the relay contact status is as follows:

- Overtemperature relay contacts: 1-3 closed, 1-4 open;
- Leakage relay contacts: 11-8 closed, 11-9 open.

If an overtemperature condition occurs, the overtemperature Red LED will turn on, the unit will turn the pump off and lock it out.

Relay contact status:

- Overtemperature relay contacts: 1-3 open, 1-4 closed;
- Leakage relay contacts: 11-8 closed, 11-9 open.

The power to the pump can be restored after the stator temperature has decreased to a point of safe operation and the thermal switches are closed. When the overtemperature condition resets, the overtemperature Red LED will turn off. The MiniCAS-120 can be reset either manually or automatically.

Note:

When selecting the "Automatic Reset" mode, the control panel should include a latching type circuit for over-temperature alarm display. This circuit will retain the information that an overtemperature situation has occurred and the operator should check the possible cause for motor overtemperature.

If a leakage is detected, after a 5 sec. delay, an alarm will be activated or the pump will be shut down and the leakage Red LED will turn on. Relay contact status:

- Overtemperature relay contacts: 1-3 closed, 1-4 open;
- Leakage relay contacts: 11-8 open, 11-9 closed.

Once the leakage condition is removed, power is restored to the pump and the Leakage Red LED will turn off automatically, leakage relay contacts will be reset.